Monopolistic competition + flexible prices ⇒ efficient allocation, with a subsidy to correct distortion of monopolistic competition

Sticky prices + policy that fully stabilizes the price level ⇒ efficient allocation

↳ Efficient allocation under social planner's problem. 由 SP 决定消费，不需要市场出清 C = Y

$$\max U(C_t, N_t)$$

s.t. $C_t(i) = A_t F(N_t(i))$ for all $i \in [0,1]$ with

$$C_t = \left(\int_0^1 C_t(i)^{\frac{\varepsilon-1}{\varepsilon}} di\right)^{\frac{\varepsilon}{\varepsilon-1}}$$

$$N_t = \int_0^1 N_t(i)\, di.$$

Optimality condition: $C_t(i) = C_t, \ \forall i \in [0,1]$ ① $N_t(i) = N_t, \ \forall i \in [0,1]$ ②

$$-\frac{U_{N,t}}{U_{C,t}} = MPN_t = (1-\alpha) A_t N_t^{-\alpha} \quad ③$$

① & ② Symmetric allocation ⇒ $MPN_t = MPN_t(i)$

Optimality conditions can be met due to: i. All goods enter utility symmetrically

... for planner, weight are the same, even distribution $C(i)$

ii. Identical technology of production

... for planner. tech for $\forall i$ the same, even distribution $N(i)$

iii. Concavity of utility.

... for planner, there's an optimal allocation of $C(i)$ & $N(i)$

$$MRS_{C,N} = -\frac{U_{N,t}}{U_{C,t}} \qquad MRT_{C,N} = \frac{dC_t}{dN_t} = F_N(N_t) \cdot A_t$$ ... output transfered one-for-one to consumption

$$MPN = A_t \cdot F_N(N_t)$$

$$\Rightarrow MRS_{C,N} = MRT_{C,N} = MPN$$

↳ Suboptimality in NK.

Two sources of distortion: ① Monopolistic competitive firms → market power.

② Sticky price

Let's study ① separately: Assume flexible price. with isoelastic demand function, optimal pricing:

$$\left.\begin{array}{l} P_t = \mu \dfrac{W_t}{MPN_t} \\ \mu = \frac{\varepsilon}{\varepsilon-1} \text{ markup} \end{array}\right\} \Rightarrow MRS_{C,N} = -\frac{U_{N,t}}{U_{C,t}} = \frac{W_t}{P_t} = \frac{MPN_t}{\mu} < MPN_t$$

Low employment & output

optimal condition ③ violated

Let τ denotes the employment subsidy. subsidy financed by lump-sum tax (no distortions)

$$P_t = \mu \cdot \frac{(1-\tau) W_t}{MPN_t} \Rightarrow -\frac{U_{N,t}}{U_{C,t}} = \frac{W_t}{P_t} = \frac{MPN_t}{\mu(1-\tau)} = MPN_t \Rightarrow \mu(1-\tau) = 1 \Rightarrow \tau = \frac{1}{\varepsilon}$$

The optimal allocation can be achieved by giving employment subsidy of $\tau = \frac{1}{\varepsilon}$

Now study ② price stickiness.

1° Average markup varies over time:

$$\mu_t = \frac{P_t}{(1-\tau)\frac{W_t}{MPN_t}} = \frac{\mu P_t}{\frac{W_t}{MPN_t}} \Rightarrow \frac{W_t}{P_t} = \frac{\mu}{\mu_t} MPN_t$$

$$-\frac{U_{N,t}}{U_{C,t}} = \frac{W_t}{P_t} = MPN_t \frac{\mu}{\mu_t}$$ If $\mu_t \neq \mu$, optimal allocation ③ violated

2° Staggered price setting: (sticky price)

$$P_t(i) \neq P_t(j), \quad C_t(i) \neq C_t(j), \quad N_t(i) \neq N_t(j)$$

Optimal condition ① and ② violated

b) Optimal Monetary Policy in the basic NK

Assumptions. 1. An optimal subsidy to offset distortion of firm's market power

2. Economy starts without relative price distortion $P_{-1}(i) = P_{-1}$ for all $i \in [0,1]$

Optimal policy should achieve: 1. $P_t = P_{t-1}$, $\forall t = 0, 1, 2 \cdots$ so that ① & ② hold

2. $\mu_t = \mu$, $\forall t$ so that ③ holds

$$\Leftrightarrow \quad \tilde{y}_t = \hat{y}_t - \hat{y}_t^n = 0, \forall t \quad \hat{\pi}_t = 0, \forall t$$

$$DIS \Rightarrow \hat{y}_t = -\frac{1}{\sigma}(\hat{i}_t - E_t[\hat{\pi}_{t+1}] - \hat{r}_t^n) + E_t[\tilde{y}_{t+1}]$$

$$\Rightarrow \quad \hat{i}_t = \hat{r}_t^n \quad \forall t.$$

1. Stablizing output: *Not freezing output at constant $\hat{y}_t = \hat{y}_t^n$ $\forall t$, $\hat{y}_t^n$ subject to technology shock.

*Eliminating gap, not eliminating fluctuations

2. Stablizing price: divine coincidence. ⇒ If central bank achieves price stability, then it automatically gets $\tilde{y}_t = 0$. Stabilizing inflation ⇒ Stabilizing output gap.

How to implement the optimal policy?

Non-policy block: DIS: $\tilde{y}_t = -\frac{1}{\sigma}(\hat{i}_t - E_t[\hat{\pi}_{t+1}] - \hat{r}_t^n) + E_t[\tilde{y}_{t+1}]$

NKPC: $\hat{\pi}_t = \beta E_t[\hat{\pi}_{t+1}] + \kappa \tilde{y}_t$ When $\hat{\pi}_t = 0$ $\forall t \Rightarrow \tilde{y}_t = 0$ $\forall t$.

Divine coincidence

Policy block:

Case a: An exogenous interest rate rule

$\hat{i}_t = \hat{r}_t^n \quad \forall t$

$\Rightarrow \begin{cases} \boxed{\tilde{y}_t} = \frac{1}{\sigma} E_t\{\hat{\pi}_{t+1}\} + E_t\{\tilde{y}_{t+1}\} \\ \hat{\pi}_t = \beta E_t\{\hat{\pi}_{t+1}\} + \kappa \tilde{y}_t \end{cases}$ plug in $\Rightarrow \begin{bmatrix} \tilde{y}_t \\ \hat{\pi}_t \end{bmatrix} = A_0 \begin{bmatrix} E_t\{\tilde{y}_{t+1}\} \\ E_t\{\hat{\pi}_{t+1}\} \end{bmatrix}$

where $A_0 = \begin{bmatrix} 1 & \frac{1}{\sigma} \\ \kappa & \beta + \frac{\kappa}{\sigma} \end{bmatrix}$

$\tilde{y}_t = \hat{\pi}_t = 0 \quad \forall t$ is one solution that associated with optimal policy, but not unique.

$\tilde{y}_t$ and $\hat{\pi}_t$ are neither pre-determined.

$\det(A_0) = \beta \quad \text{trace}(A_0) = 1 + \beta + \frac{\kappa}{\sigma}$.

$p(1) = 1 - T + D = 1 - 1 - \beta - \frac{\kappa}{\sigma} + \beta = -\frac{\kappa}{\sigma} < 0$.

$p(-1) = 1 + T + D = 1 + 1 + \beta + \frac{\kappa}{\sigma} + \beta = 2 + 2\beta + \frac{\kappa}{\sigma} > 0$

There is one root between $(-1, 1)$ and another root outside unit circle.

$\Rightarrow$ # jump variable. > # unstable root $\Rightarrow$ System is indeterminated

* No guarantee on the realisation of $\tilde{y}_t = \hat{\pi}_t = 0, \ \forall t$

Case b: a Taylor-type interest rate rule.

$\hat{i}_t = \hat{r}_t^n + \phi_\pi \hat{\pi}_t + \phi_y \tilde{y}_t$ where $\phi_\pi \geq 0, \phi_y \geq 0 \quad \forall t$.

$\Rightarrow \begin{cases} \tilde{y}_t = -\frac{\phi_\pi}{\sigma}\hat{\pi}_t - \frac{\phi_y}{\sigma}\tilde{y}_t + \frac{1}{\sigma} E_t\{\hat{\pi}_{t+1}\} + E_t\{\tilde{y}_{t+1}\} \\ \hat{\pi}_t = \beta E_t\{\hat{\pi}_{t+1}\} + \kappa \tilde{y}_t. \end{cases}$

$\Rightarrow \begin{cases} \tilde{y}_t = \frac{1 - \phi_\pi \beta}{\sigma + \phi_y + \kappa\phi_\pi} E_t\{\hat{\pi}_{t+1}\} + \frac{\sigma}{\sigma + \phi_y + \kappa\phi_\pi} E_t\{\tilde{y}_{t+1}\} \\ \hat{\pi}_t = \frac{\kappa + \sigma\beta + \beta\phi_y}{\sigma + \phi_y + \kappa\phi_\pi} E_t\{\hat{\pi}_{t+1}\} + \frac{\sigma\kappa}{\sigma + \phi_y + \kappa\phi_\pi} E_t\{\tilde{y}_{t+1}\} \end{cases}$

$\Rightarrow \begin{bmatrix} \tilde{y}_t \\ \hat{\pi}_t \end{bmatrix} = A_T \begin{bmatrix} E_t[\tilde{y}_{t+1}] \\ E_t[\hat{\pi}_{t+1}] \end{bmatrix}$ where $A_T = \Omega \begin{bmatrix} \sigma & 1 - \beta\phi_\pi \\ \sigma\kappa & \kappa + \beta(\sigma + \phi_y) \end{bmatrix}$

$\Omega = \frac{1}{\sigma + \phi_y + \kappa\phi_\pi}$

$\tilde{y}_t = \hat{\pi}_t = 0 \ \forall t$ is always a solution. To guarantee uniqueness, we need two stable roots (notice that $x_t = A_T E_t x_{t+1}$, but BK is defined on $E_t x_{t+1} = A_T^{-1} x_t$, thus 2 stable roots $\Rightarrow$ 2 unstable roots in A_T^{-1})

$$\text{trace}(A^T) = \Omega\,[\sigma + \kappa + \beta(\sigma + \phi_y)]$$

$$\det(A^T) = \Omega^2 \det(M) = \Omega\beta\sigma$$

To have stable result, need $D<1$, $p(1)>0$, $p(-1)>0$

$$p(1) = 1 - tr(A^T) + \det(A^T) = 1 - \Omega[\sigma + \kappa + \beta\phi_y] > 0$$

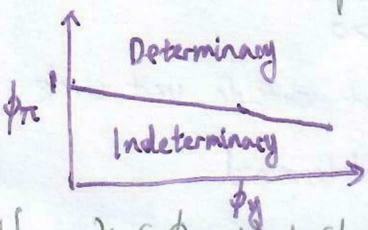
$$\Rightarrow \sigma + \kappa + \beta\phi_y < \sigma + \phi_y + \kappa\phi_\pi$$

$$\Rightarrow \kappa(\phi_\pi - 1) + (1-\beta)\phi_y > 0 \quad (*)$$

$$p(-1) = 1 + tr(A^T) + \det(A^T) = 1 + \Omega[\sigma + \kappa + \beta\phi_y + \beta\sigma] > 0$$ for sure

$$\Delta = Tr(A^T)^2 - 4\cdot\det(A^T) = \Omega^2[\sigma + \kappa + \beta(\sigma + \phi_y)]^2 - 4\Omega\beta\sigma.$$ sign does not matter.

$$D = \det(A^T) = \Omega\beta\sigma = \frac{\beta\sigma}{\sigma + \phi_y + \kappa\phi_\pi} = \frac{1}{\frac{1}{\beta} + \frac{\phi_y + \kappa\phi_\pi}{\beta\sigma}} \overset{(*)\ \&\ \beta\in(0,1)}{<} \frac{1}{1 + \frac{\kappa - \kappa\phi_\pi + \kappa\phi_\pi}{\beta\sigma}}$$



$$= \frac{1}{1 + \frac{\kappa}{\beta\sigma}} < 1$$

If $\kappa(\phi_\pi - 1) + (1-\beta)\phi_y > 0$ is satisfied. $\tilde{y}_t = \hat{\pi}_t = 0$ and $\hat{i}_t = \hat{r}_t^n\ \forall t$ hold

If $\phi_y = 0 \Rightarrow \kappa(\phi_\pi - 1) > 0 \Rightarrow \phi_\pi > 1$ Taylor principle

If $\phi_y > 0 \Rightarrow$. Output gap helps, lower bound on ϕ_π can be relaxed a little.

but $1-\beta$ is usually small, inflation is the main stabilising force.

Case c: A forward-looking interest rate rule

$$\hat{i}_t = \hat{r}_t^n + \phi_\pi E_t[\hat{\pi}_{t+1}] + \phi_y E_t[\tilde{y}_{t+1}]$$

$$\Rightarrow \begin{cases} \tilde{y}_t = -\frac{\phi_\pi - 1}{\sigma} E_t[\hat{\pi}_{t+1}] - \frac{\phi_y - \sigma}{\sigma} E_t[\tilde{y}_{t+1}] \\ \hat{\pi}_t = \beta E_t[\hat{\pi}_{t+1}] + \kappa \tilde{y}_t \quad (*) \end{cases}$$

$$\Rightarrow \begin{cases} \tilde{y}_t = \frac{1-\phi_\pi}{\sigma} E_t[\hat{\pi}_{t+1}] + \frac{\sigma - \phi_y}{\sigma} E_t[\tilde{y}_{t+1}] \quad (**) \\ \hat{\pi}_t = \frac{\kappa(1-\phi_\pi) + \beta\sigma}{\sigma} E_t[\hat{\pi}_{t+1}] + \frac{\kappa(\sigma - \phi_y)}{\sigma} E_t[\tilde{y}_{t+1}] \quad (***) \end{cases}$$

$$\Rightarrow \begin{bmatrix} \tilde{y}_t \\ \hat{\pi}_t \end{bmatrix} = A_F \begin{bmatrix} E_t[\tilde{y}_{t+1}] \\ E_t[\hat{\pi}_{t+1}] \end{bmatrix}, \quad \text{where } A_F = \begin{bmatrix} \frac{\sigma - \phi_y}{\sigma} & \frac{1-\phi_\pi}{\sigma} \\ \frac{\kappa(\sigma - \phi_y)}{\sigma} & \beta + \frac{\kappa(1-\phi_\pi)}{\sigma} \end{bmatrix}$$

$$\text{trace}(A_F) = \beta + \frac{\kappa(1-\phi_\pi) + (\sigma - \phi_y)}{\sigma} \qquad \det(A_F) = \beta\,\frac{\sigma - \phi_y}{\sigma}$$

$p(1) = 1 - tr(A_F) + det(A_F) = 1 - \beta - \frac{k(1-\phi_\pi) + (\sigma - \phi_y)}{\sigma} + \beta \frac{\sigma - \phi_y}{\sigma} > 0$

$\Rightarrow 1 - \beta - \frac{k}{\sigma}(1-\phi_\pi) - 1 + \frac{1}{\sigma}\phi_y + \beta - \frac{\beta}{\sigma}\phi_y > 0$

$\Rightarrow k(\phi_\pi - 1) + (1-\beta)\phi_y > 0$ (1) ... policy cannot be too passive

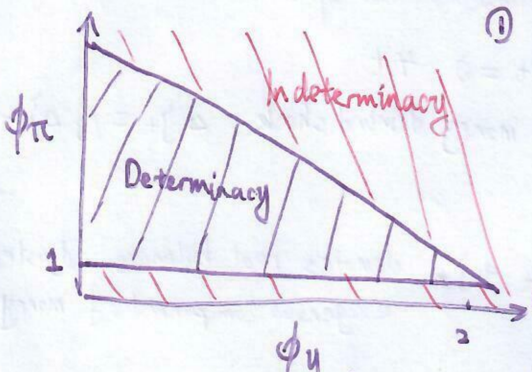
$p(-1) = 1 + tr(A_F) + det(A_F) = 1 + \beta + \frac{k}{\sigma}(1-\phi_\pi) + 1 - \frac{1}{\sigma}\phi_y + \beta - \frac{\beta}{\sigma}\phi_y > 0$

$\Rightarrow k(\phi_\pi - 1) + (1+\beta)\phi_y < 2\sigma(1+\beta)$ (2) ... policy cannot be too agressive.

$D < 1 \Rightarrow \beta(1 - \frac{\phi_y}{\sigma}) < 1$ since $\beta \in (0,1)$ & $\phi_y \geq 0$, so this hold.

$\Rightarrow$ Under (1) & (2), we have 2 stable roots in A_F, 2 unstable roots in A_F^{-1}

$\tilde{y}_t = \hat{\pi}_t = 0 \quad \forall t$ is the unique solution.



① (**) if $\phi_\pi > 1$, then coefficient on $E_t \hat{\pi}_{t+1}$ is negative.

$E_t[\hat{\pi}_{t+1}] \uparrow \Rightarrow \tilde{y}_t \downarrow \Rightarrow \hat{\pi}_t \downarrow$ (*)

If $\phi_\pi < 1 \Rightarrow E_t[\hat{\pi}_{t+1}] \uparrow \Rightarrow \tilde{y}_t \uparrow \Rightarrow \hat{\pi}_t \uparrow$

selffulfilling inflation, policy can not be too passive

② If policy is too agressive so that (2) is violated,

$p(-1) < 0$, one eigenvalue crosses -1, system unstable.

But All optimal policy listed are not realistic, since policy rate be adjusted one-for-one with the natural interest rate. But $\hat{r}_t^n$ is usually unobservable, because it depends on the true model of the economy, parameter values & the realized shocks. $\Rightarrow$ Simple rules are needed!

Due to divine coincidence, unobservable output gap is not the main issue, since $E_t \hat{\pi}_{t+1} = 0 \Rightarrow \tilde{y}_t = 0$

9) Simple monetary policy rule.

① Rotemberg and Woodford (1999): Second-order approximation to the utility losses due to deviation from efficient allocation.

$$W = \frac{1}{2} E_0 \sum_{t=0}^{\infty} \beta^t \left[(\sigma + \frac{\varphi + \alpha}{1-\alpha}) \tilde{y}_t^2 + \frac{\varepsilon}{\lambda} \hat{\pi}_t^2 \right]$$

Per-period Loss: $L = \frac{1}{2}\left[(\sigma + \frac{\varphi+\alpha}{1-\alpha}) Var(\tilde{y}_t) + \frac{\varepsilon}{\lambda} Var(\hat{\pi}_t) \right]$

$\sigma \uparrow, \varphi \uparrow, \alpha \uparrow \Rightarrow L \uparrow$ curvatures amplify the effect (consumption, labor, tech)

$\varepsilon \uparrow, \frac{1}{\lambda} \uparrow \Rightarrow L \uparrow$ ε amplify losses from price dispersion, $\frac{1}{\lambda}$ amplify the degree of price dispersion

② Taylor-type interest rate rule: $\hat{i}_t = \phi_\pi \hat{\pi}_t + \phi_y \hat{y}_t$, $\phi_\pi > 0$, $\phi_y > 0 \Rightarrow \hat{i}_t = \phi_\pi \hat{\pi}_t + \phi_y \tilde{y}_t + \underbrace{\phi_y \hat{y}_t^n}_{V_t}$

Plug in $\Rightarrow$
$$\begin{cases} DIS \Rightarrow (\sigma + \phi_y)\tilde{y}_t + \phi_\pi \hat{\pi}_t = \sigma E_t \tilde{y}_{t+1} + E_t \hat{\pi}_{t+1} + (\hat{r}_t^n - V_t) \\ \hat{\pi}_t = \beta E_t \hat{\pi}_{t+1} + \lambda \tilde{y}_t \end{cases}$$

$\Rightarrow \begin{bmatrix} \tilde{y}_t \\ \hat{\pi}_t \end{bmatrix} = A_T \begin{bmatrix} E_t \tilde{y}_{t+1} \\ E_t \hat{\pi}_{t+1} \end{bmatrix} + B_T (\hat{r}_t^n - V_t)$ $A_T = \Omega \begin{bmatrix} \sigma & 1 - \beta\phi_\pi \\ k\sigma & k + \beta(\sigma + \phi_y) \end{bmatrix}$, $B_T = \Omega \begin{bmatrix} 1 \\ k \end{bmatrix}$, $\Omega = \frac{1}{\sigma + \phi_y + k\phi_\pi}$

$\hat{a}_t = \rho_a \hat{a}_{t-1} + \varepsilon_t^a$, $\hat{y}_t^n = \psi_{ya}\hat{a}_t$ where $\psi_{ya}^n = \frac{1+\phi}{\sigma+\phi+\alpha(1-\sigma)} > 0$

$\Rightarrow v_t = \phi_y \hat{y}_t^n = \phi_y \psi_{ya}^n \hat{a}_t$ $\qquad \hat{r}_t^n = \sigma E_t\{\Delta \hat{y}_{t+1}^n\} \Rightarrow \hat{r}_t^n = -\sigma \psi_{ya}^n (1-\rho_a)\hat{a}_t$

$\Rightarrow \hat{r}_t^n - v_t = -\psi_{ya}^n[\sigma(1-\rho_a) + \phi_y]\hat{a}_t$ $\qquad \Delta\hat{y}_{t+1}^n = \hat{y}_{t+1}^n - \hat{y}_t^n = \psi_{ya}^n(\hat{a}_{t+1} - \hat{a}_t)$

$E_t \Delta \hat{y}_{t+1}^n = \psi_{ya}^n(\rho_a - 1)\hat{a}_t$

$$B_T(\hat{r}_t^n - v_t) = \begin{bmatrix} 1 \\ \kappa \end{bmatrix} \Omega \{-\psi_{ya}^n[\sigma(1-\rho_a)+\phi_y]\}\hat{a}_t$$

shock loading: $f(\phi_y) = \frac{\sigma(1-\rho_a)+\phi_y}{\sigma+\phi_y+\kappa\phi_\pi} \Rightarrow f'(\phi_y) = \frac{\sigma\rho_a + \kappa\phi_\pi}{(\sigma+\phi_y+\kappa\phi_\pi)^2} > 0$

Shock term gets larger if the central bank responds more agresively to output

$\Rightarrow$ output gap and inflation variance $\uparrow$ welfare $\downarrow$

$\phi_\pi \to +\infty \Rightarrow f(\phi_y) \to 0 \Rightarrow$ the exogenous driving force hitting the system goes to 0

$\Rightarrow$ Aggressive inflation targeting gets arbitrarily close to the optimum

2) A constant money growth rule Friedman (1960): $\Delta \hat{m}_t = 0 \ \forall t$

$\underbrace{\hat{m}_t - \hat{p}_t}_{\hat{m}} = \hat{y}_t - \eta \hat{i}_t - \zeta_t$, ζ_t is an exogenous money demand shock: $\Delta\zeta_t = \rho_\zeta \Delta\zeta_{t-1} + \varepsilon_t^\zeta$

$\hat{m}_t = \tilde{y}_t + \hat{y}_t^n - \eta \hat{i}_t - \zeta_t$ $\qquad$ Let $\hat{m}_t^+ = \hat{m}_t + \zeta_t$ denotes real balances adjusted by exogenous component of money demand

$\Rightarrow \hat{i}_t = \frac{1}{\eta}(\tilde{y}_t + \hat{y}_t^n - \hat{m}^+)$

$\hat{m}_t^+ - \hat{m}_{t-1}^+ = \underbrace{\hat{m}_t - \hat{m}_{t-1}}_{0} \underbrace{- \hat{p}_t + \hat{p}_{t-1}}_{\hat{\pi}_t} + \zeta_t - \zeta_{t-1} \Rightarrow \hat{m}_{t-1}^+ = \hat{m}_t^+ + \hat{\pi}_t - \Delta\zeta_t$

$$A_{M,0}\begin{bmatrix} \tilde{y}_t \\ \hat{\pi}_t \\ \hat{m}_{t-1}^+ \end{bmatrix} = A_{M,1}\begin{bmatrix} E_t\{\tilde{y}_{t+1}\} \\ E_t\{\hat{\pi}_{t+1}\} \\ \hat{m}_t^+ \end{bmatrix} + B_M \begin{bmatrix} \hat{r}_t^n \\ \hat{y}_t^n \\ \Delta\zeta_t \end{bmatrix}$$

** The degree of stability of money demand is a key element in determining the desirability of a money growth rule.